Does Shrinking a CNN Make It Easier to Fool with a Sticker?

Model compression and adversarial patch robustness on CIFAR-10: hypotheses, method, results and what they mean

A plain-language companion to the paper draft. Every number below is taken from the generated result tables in paper/generated/ (3 training seeds: 42, 43, 44) or, where stated, from the seed-42 log in PROGRESS.md.

In one paragraph. Compression is not one thing. **Quantization and pruning**, which keep the original network's weights, leave patch robustness and certified robustness essentially unchanged, even at about 8× smaller storage. **Retraining a smaller network**, either from scratch (dense-small) or by distillation (KD), costs robustness, and the cost grows with the compression ratio: at 8×, defended certified accuracy against a 5 px patch falls by 10.4 points for dense-small and 7.7 points for KD. Pruned and dense-small models with the same number of non-zero weights differ by up to 9.8 certified points, so the damage comes from **lost capacity, not sparsity**. Patches made on the full model carry over almost fully to its quantized copies, which matters for anyone who deploys a quantized version of public weights.

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1. Background

An **adversarial patch** is a small, visible square of pixels that, once pasted anywhere on an image, makes a classifier output the wrong label. Unlike invisible noise attacks, a patch can be printed as a sticker, so it is the attack that matters for cameras in the physical world. Two kinds are studied here:

- **Universal patch** (Brown et al.): one patch, trained once, that pushes *any* image towards a chosen target class. We train one per target class (all 10) so no target is cherry-picked.
- **Per-image patch** (LaVAN): a patch optimised separately for each image. This is the stronger, worst-case attacker.

Model compression makes a network small enough for phones and embedded devices. The three standard families work very differently:

- **Quantization** stores the same weights at lower precision (FP16, INT8, INT4).
- **Pruning** sets the smallest weights to zero (50%, 75%, 87.5%) and fine-tunes the rest.
- **Knowledge distillation (KD)** trains a new, narrower network to imitate the full one.

PatchCleanser (Xiang et al., 2022) is a *certified* defence: it masks out regions of the image and, if every masked prediction agrees, it can *prove* that no patch of a given size anywhere in the image could change the answer. Certified accuracy is therefore a guarantee, not just a measurement against one attack.

The gap. Closest prior work studied invisible universal perturbations (not patches) on pruned and quantized models, and PatchCleanser was evaluated only on full-size classifiers. Nobody had asked whether a deployed, compressed model keeps the patch robustness, and the certificates, of the model it came from.

2. Research questions and hypotheses

The work plan fixed four research questions. For each, the table states the hypothesis we set out to test: the common intuition a reader would bring, written so the data can contradict it.

	Research question	Hypothesis tested
RQ1	How do pruning, quantization and distillation (2×, 4×, 8×) change vulnerability to universal and per-image patches?	**H1.** Compression removes redundancy, so every family becomes more vulnerable, and more so at higher ratios.
RQ2	Do patches transfer between the full model and its compressed versions, in both directions?	**H2.** Compressed models are derived from the full model, so patches transfer well between them in both directions.
RQ3	How much PatchCleanser certified accuracy survives once the base classifier is compressed?	**H3.** Certified accuracy falls with compression for every family.
RQ4	Are pruning's effects due to sparsity, or to the lost capacity?	**H4.** A pruned model behaves like a dense model with the same number of non-zero weights, so sparsity itself is what matters.

The research plan phrased these as open questions. H1–H4 are the 'default expectation' versions, used here so each result can be read as confirming, refuting or refining a specific claim.

3. How the study was built

3.1 Models: one baseline, twelve compressed versions

The baseline is a CIFAR-10 ResNet-18 (11.17 M parameters, about 95.2% clean accuracy). Every compression family is run at nominal 2×, 4× and 8× size reduction, giving 12 compressed models:

- **Quantization:** FP16, INT8 (post-training), INT4 (quantization-aware training).
- **Pruning:** global unstructured magnitude pruning at 50 / 75 / 87.5% sparsity, 40-epoch fine-tune.
- **Dense-small (control):** a narrower ResNet-18 (width 45 / 32 / 23) trained from scratch, with the same parameter count as the pruned model's non-zero weights. This control is what lets RQ4 separate sparsity from capacity.
- **Distillation:** the same narrow architectures trained to imitate the baseline (Hinton KD, T = 4).

Matching sizes across families is a known trap. Models are matched on the *nominal* ratio and the real *storage* ratio is reported alongside, e.g. 87.5% pruning stores at 6.37× rather than 8× once the sparsity mask is counted (Table A1).

3.2 Attacks

- Patch sides of 5 px (2.4% of the image) and 8 px (6.25%), pasted at random locations.
- **Universal:** 10 targeted patches per model, trained with location randomisation (EOT), evaluated on the full 10k test set at fixed seeded locations so every model sees the same placements.
- **LaVAN:** per-image patches on a fixed 1,000-image subset, 100 steps × 10 restarts at new locations.
- **A noise-patch control** (random pixels, same size) measures how much damage is plain occlusion rather than adversarial optimisation.
- **Quantized models are attacked twice**, with straight-through gradients and with gradients from an FP32 copy (BPDA), and the stronger attack is reported. This rules out 'robustness' that is really only broken gradients.

3.3 Defence: a defended model zoo

On the vanilla baseline PatchCleanser certifies only 11.0% (5 px) and 1.9% (8 px) of images. Comparing compression families on numbers that close to zero would say nothing. PatchCleanser's authors also train with masks, so we retrained the baseline with **mask augmentation** and rebuilt all twelve compressed models from it with augmentation kept on. The order is **defend, then compress**, which is how a deployed model is built. This lifted baseline certified accuracy to 71.6% (5 px) and 53.6% (8 px).

3.4 Statistics

The full pipeline (13 models → attacks → transfer → certification, vanilla and defended) was run for three training seeds (42, 43, 44), 78 trained models in total. Results are mean ± standard deviation over seeds. Each comparison is a *paired* difference per seed (the attack seed is shared, so the pairing is real), with a 95% t-interval (n = 3, df = 2). **A difference is called real only when its interval excludes zero.** With three seeds the intervals are wide, so this is a conservative rule. There is no multiple-comparison correction, which the paper states.

4. Results

4.1 RQ1: Patch vulnerability across compression families

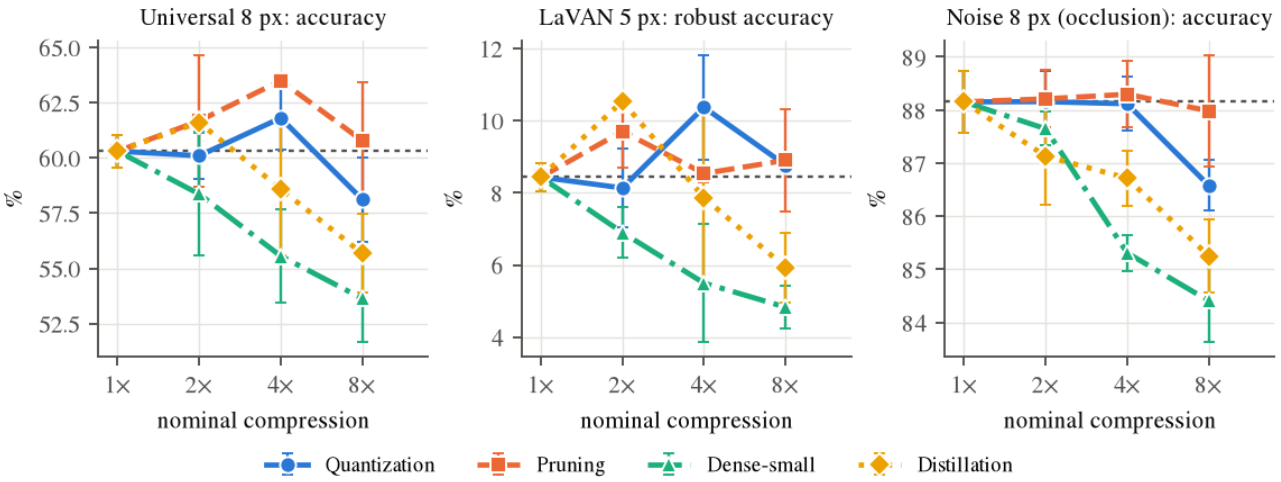


Figure 1. Accuracy under attack vs. nominal compression (mean ± std over 3 seeds; the dashed line is the baseline). Left: universal 8 px patch. Middle: LaVAN 5 px per-image patch. Right: random-noise 8 px patch, i.e. occlusion alone. Higher is more robust.

What the figure shows. Quantization and pruning stay on or above the baseline line at every ratio. Dense-small falls steadily as it shrinks (universal 8 px accuracy 60.3 → 58.4 → 55.6 → 53.6%), and KD holds at 2× but drops by 8× (55.7%). The right panel shows that the same two families also lose accuracy to a *random* patch (dense-small 8×: 84.4% vs. 88.2%). Part of their weakness is plain sensitivity to occlusion, before any adversarial optimisation.

Which differences survive the statistics. Versus the baseline, universal 8 px accuracy drops significantly for dense-small 4× (−4.7 pp), dense-small 8× (−6.7 pp) and KD 8× (−4.6 pp), and LaVAN robust accuracy drops for dense-small 8× (−3.6 pp) and KD 8× (−2.5 pp). No quantized model differs significantly from the baseline on any attack metric. Pruning 4× is significantly *better* on universal 8 px accuracy (+3.2 pp), and KD 2× on LaVAN (+2.1 pp).

Two caveats. LaVAN with an 8 px patch fools every model (robust accuracy close to 0%, targeted hit rates of 96–98%), so only the 5 px setting separates models. The INT4 dry runs found the straight-through attack slightly *stronger* than the FP32 surrogate, so quantization's neutral result is not caused by gradient masking.

4.2 RQ2: Do patches transfer?

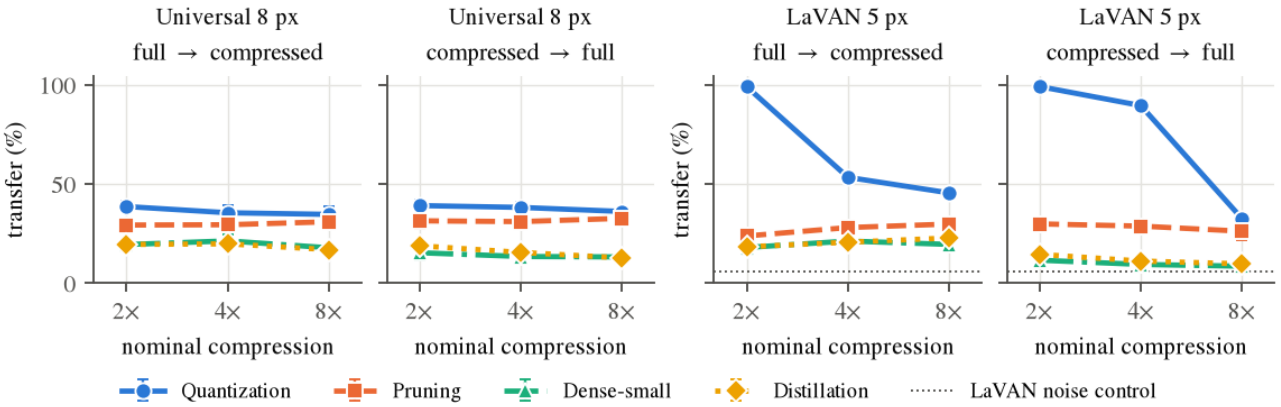


Figure 2. Transfer between the full model and each compressed model, in both directions (mean ± std over 3 seeds). Universal: target hit rate of a patch made on one model and pasted on the other. LaVAN: share of images fooled, counted over images both models get right and the source patch fooled. The dotted line is the noise-patch control.

Transfer separates the four families into three groups:

- **Quantization keeps the full model's attack surface.** FP16 is the same model to an attacker (99% LaVAN transfer both ways). Universal patches from the full model keep most of their power on INT8 / INT4 (35.4 / 34.6% hit vs. 37.1 / 41.8% white-box). Per-image LaVAN patches from the full model fool INT8 53% and INT4 45% of the time, against a 6–7% noise control.
- **Pruning is in the middle and roughly symmetric:** universal hit about 29–33% in both directions, LaVAN 24–30%.
- **Retrained small models are the most isolated, and the transfer is asymmetric.** Patches go from full to small (universal 17–21%) much better than from small to full (13–19%). For dense-small, LaVAN small → full (8.5–11.5%) is close to the noise control (7.6–8.1%).

Within quantization there is a striking asymmetry: LaVAN patches made on INT8 fool the full model 89.5% of the time, while the reverse reaches only 53.3%. Per-image patches stop at the first success, so they sit right on the decision boundary, and a quantization bin is enough to push a full-model patch back across it. Patches made on the coarser model carry a margin that survives the trip to the smooth model.

KD students sit slightly closer to the teacher than dense-small models of the same size (universal compressed → full: 18.8 / 15.5 / 12.7% vs. 15.3 / 13.4 / 13.2%), consistent with distillation copying part of the teacher's decision features. The gap closes at 8×.

4.3 RQ3: Certified robustness under compression

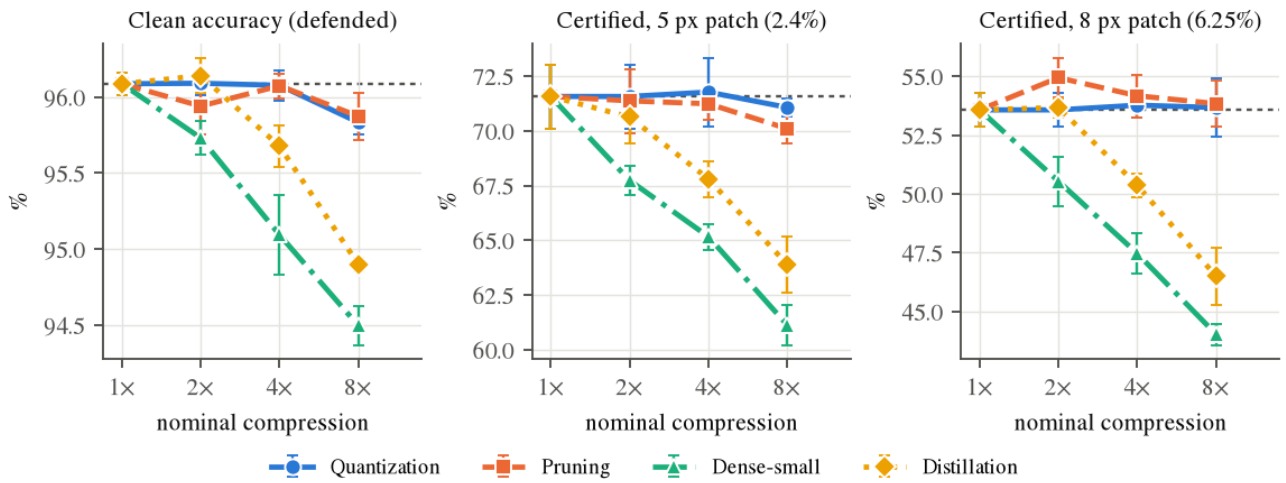


Figure 3. Defended model zoo (mask-augmented, defend-then-compress). Left: clean accuracy. Middle / right: PatchCleanser certified accuracy against a 5 px and an 8 px patch. Mean ± std over 3 seeds; the dashed line is the defended baseline.

This is the central result. With the defended zoo, certified accuracy is flat for quantization (INT8 71.8 / 53.8%, INT4 71.1 / 53.7% vs. baseline 71.6 / 53.6%) and for pruning (8×: 70.1 / 53.8%). It falls steeply for the retrained small models: dense-small 8× certifies 61.1 / 44.0% and KD 8× 63.9 / 46.5%. Paired against the baseline, the 5 px certified loss is −10.4 pp for dense-small 8× and −7.7 pp for KD 8×. The 8 px loss is −9.5 and −7.1 pp. All four intervals exclude zero.

Certified accuracy falls much faster than clean accuracy. Dense-small 8× loses only 1.6 points of defended clean accuracy (94.5 vs. 96.1%) but 10.5 points of certified accuracy. A model that looks almost as good on the standard benchmark can carry a much weaker guarantee. For the vanilla (undefended) models, every family certifies about 8–11% (5 px) and 1–2% (8 px), confirming that certification is only informative with a defended base model.

4.4 RQ4: Sparsity or capacity?

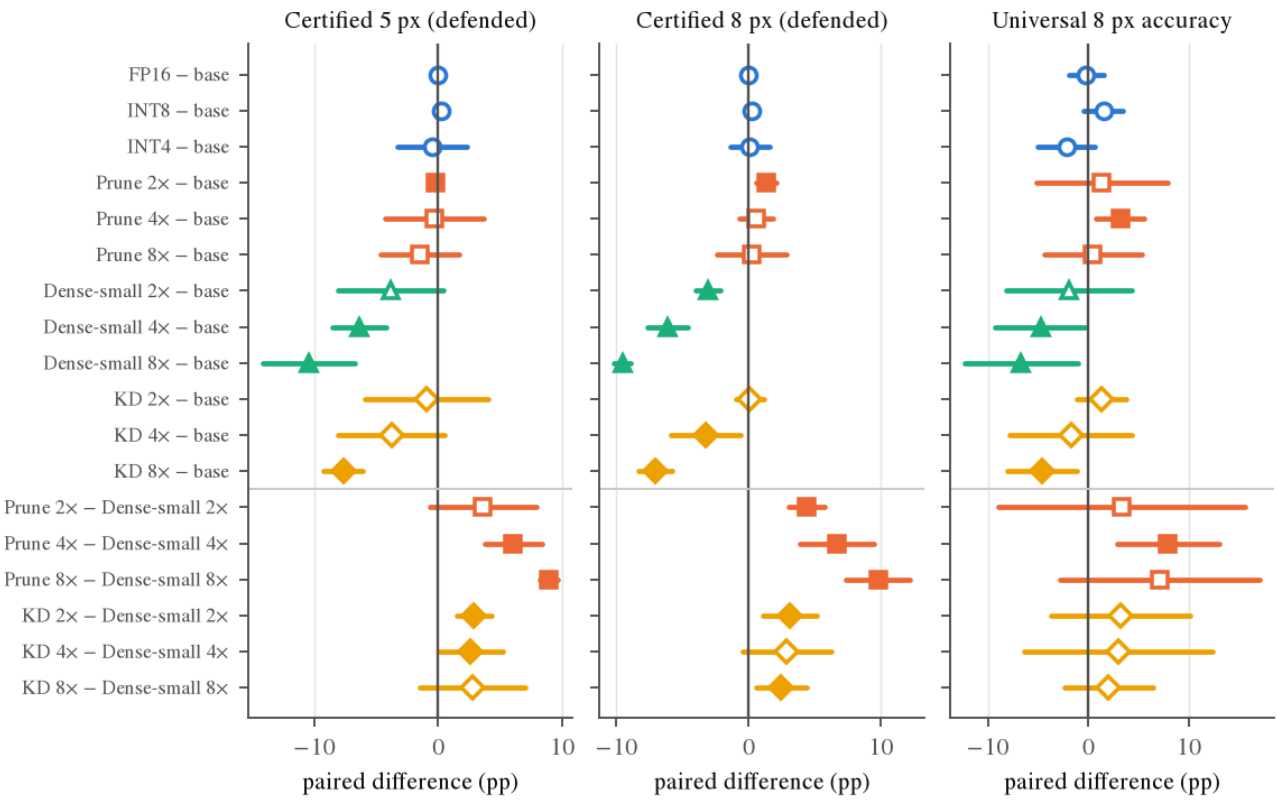


Figure 4. Paired differences in percentage points with 95% t-intervals over 3 seeds. Filled markers = interval excludes zero. Top block: each compressed model minus the baseline. Bottom block: pruned minus dense-small, and KD minus dense-small, at equal parameter count.

The bottom block of Figure 4 answers RQ4 directly. A pruned network and a dense-small network with the same number of non-zero weights are *not* equivalent. The pruned model certifies significantly more at every ratio where the interval resolves: +6.1 / +6.7 pp (5 / 8 px) at 4× and +9.0 / +9.8 pp at 8×. It is also +7.9 pp more accurate under the universal 8 px patch at 4×. The gap *grows* with compression. KD recovers part of the loss relative to dense-small (+2.5 to +3.6 pp on the robustness metrics where the interval excludes zero) but not all of it.

5. Deductions: from evidence to conclusion

Each hypothesis is set against the evidence and judged. 'Refuted' means the data contradict the hypothesis as stated; 'refined' means it holds only for part of the design space.

H1 Every compression family becomes more vulnerable to patches, more so at higher ratios. **Refuted (as stated)**

Evidence. No quantized model differs significantly from the baseline on universal or LaVAN accuracy, and INT8 / pruned models are nominally *more* robust (pruning 4× significantly so, +3.2 pp). Only dense-small (4×, 8×) and KD (8×) are significantly worse, and their loss grows with the ratio.

Therefore. Vulnerability depends on *how* a model is compressed, not on how much. Methods that keep the original weights (quantize, prune + fine-tune) preserve robustness. Methods that retrain a smaller network lose it, in proportion to the capacity removed.

H2 Patches transfer well between the full model and its compressed versions, in both directions. **Refined**

Evidence. True for quantization (FP16 99%; universal full → INT8/INT4 keeps most of its hit rate; LaVAN 45–53% vs. a 6–7% noise floor). Partial for pruning (24–33%). Weak and one-directional for dense-small and KD, where small → full LaVAN transfer is close to the noise floor.

Therefore. Transfer tracks how much of the original network survives compression. The practical threat is the quantized case: an attacker who holds public FP32 weights can attack a quantized deployment almost as well as the deployed model itself, without ever touching it.

H3 Certified accuracy falls with compression for every family. **Refined**

Evidence. Flat for quantization (all within ±0.5 pp, intervals cover zero) and pruning (one small significant change of each sign at 2×). Large, significant falls for dense-small (−3.0 to −10.4 pp) and KD (−3.2 to −7.7 pp), growing with the ratio, and much larger than their clean-accuracy losses.

Therefore. PatchCleanser's guarantee survives weight-preserving compression up to about 8× storage reduction. It does not survive capacity reduction. Clean accuracy is a poor proxy for how much certified robustness was lost.

H4 Pruned models behave like dense models with the same number of non-zero weights. **Refuted**

Evidence. At equal non-zero parameters, pruned beats dense-small on certified accuracy by +6.1 to +9.8 pp at 4× and 8× (intervals exclude zero), on clean accuracy by +0.8 to +1.4 pp, and on LaVAN robust accuracy at 2× and 8×. The gap widens as compression increases.

Therefore. Sparsity is not the cause. Lost capacity is. A pruned network inherits the full model's learned features, including whatever makes it robust to occlusion and to the defence's masks. A small network trained from scratch never learns them, and distillation transfers only part of them.

Putting the four together

The four deductions point to one organising variable: **how much of the full model's learned function the compressed model keeps**. Ranked by that variable (FP16 > INT8 > INT4 ≈ pruning > KD > dense-small), the families line up in the same order on robustness, certification and transfer. The weakness of retrained small models also shows up with a *random* noise patch (Figure 1, right), so part of the mechanism is a general sensitivity to occluded pixels, which is exactly what PatchCleanser's masks introduce. That is a

plausible reason certification suffers most.

> **Status of the 'why'.** The J8 analysis that would test this mechanism directly (occlusion sensitivity, input-gradient concentration, CKA similarity and prediction agreement with the baseline, correlated with vulnerability and transfer) is not finished. Its run stopped after 66 of 78 models and the correlation summary is still empty. The mechanism above is therefore an *interpretation* consistent with the results, not yet a measured finding.

6. Discussion

6.1 Practical guidance

- **To shrink a patch-defended model, prefer quantization (down to INT4) or unstructured pruning (up to 87.5%).** Both kept certified accuracy within noise of the full model at up to about 8× storage reduction.
- **Avoid replacing a defended model with a smaller retrained one** (dense or distilled) without re-measuring certified accuracy. The loss can be 7–10 points at 8× while clean accuracy moves by about 1 point.
- **If distillation is required, it is better than training the small model from scratch**, but it does not close the gap.
- **Treat a quantized deployment of public weights as already exposed.** Patches made on the FP32 model transfer to it, so security cannot rest on keeping the quantized model private.
- **Report certified accuracy, not clean accuracy, when comparing compressed defended models.**

6.2 Limitations and threats to validity

- **One dataset, one architecture.** CIFAR-10 at 32 px with ResNet-18. The planned GTSRB, ImageNette (224 px, ResNet-50) and MobileNetV2 settings were dropped for the submission deadline. Patch-to-image ratios and redundancy differ at higher resolution, so generality is not established.
- **Three seeds.** t-intervals with df = 2 are wide, so many real but small effects (e.g. INT4 −2.2 pp on universal 8 px) are not resolved. No multiple-comparison correction is applied.
- **Quantization is simulated** (fake-quant with BN folding). Integer kernels on real hardware may differ slightly. Latency numbers are only meaningful for dense models.
- **Unstructured pruning only.** Structured (channel) pruning would yield a narrow dense model, i.e. it would behave like dense-small, which our results suggest is the harmful case. That prediction is untested.
- **Defend-then-compress only.** Compress-then-defend (fine-tuning each compressed model with masks) might recover some of the dense-small / KD loss.
- **LaVAN transfer is a lower bound.** Per-image patches stop at the first success, sit on the decision boundary and transfer worse than high-margin patches would. Seed-42 LaVAN patches were also stored at FP16 precision, which lowers their replay success on quantized models (fixed for seeds 43 / 44).
- **Physical robustness is a single-seed ablation.** With rotation and brightness changes all 4× models stay within about 3 points of each other (seed 42, Table A4).

6.3 What is still open

- Finish J8 (12 models left) and run the correlation analysis to test the occlusion-sensitivity / feature-retention explanation quantitatively.
- A second setting (GTSRB or a second architecture) to check that the quantize/prune vs. retrain split holds beyond CIFAR-10.
- Structured pruning and compress-then-defend ablations at 4×.
- A LaVAN confidence-margin ablation to measure how much stronger transfer becomes with high-confidence patches.

Appendix: Full result tables

All values in %, mean ± std over seeds 42–44 unless noted. Tables are rendered directly from paper/generated/*.tex.

Table A1. Models

Model	Storage	Nonzero	Clean (vanilla)	Clean (defended)
Baseline	1.00×	11.17M	95.2 ±0.1	96.1 ±0.1
FP16	2.00×	11.17M	95.2 ±0.1	96.1 ±0.1
INT8	4.00×	11.17M	95.2 ±0.0	96.1 ±0.1
INT4	7.98×	11.17M	95.0 ±0.1	95.8 ±0.1
Prune 2×	1.88×	5.59M	95.3 ±0.1	95.9 ±0.2
Prune 4×	3.55×	2.80M	95.4 ±0.1	96.1 ±0.1
Prune 8×	6.37×	1.41M	95.4 ±0.1	95.9 ±0.2
Dense-small 2×	2.02×	5.53M	95.2 ±0.0	95.7 ±0.1
Dense-small 4×	3.99×	2.80M	94.6 ±0.2	95.1 ±0.3
Dense-small 8×	7.72×	1.45M	94.0 ±0.1	94.5 ±0.1
KD 2×	2.02×	5.53M	95.3 ±0.2	96.1 ±0.1
KD 4×	3.99×	2.80M	94.9 ±0.2	95.7 ±0.1
KD 8×	7.72×	1.45M	94.4 ±0.2	94.9 ±0.0

Storage = ratio of stored size vs. FP32 baseline (pruned models include a 1-bit mask). Nonzero = non-zero parameters. Clean accuracy on the 10k test set.

Table A2. Patch attacks (RQ1)

Model	U5 acc	U8 acc	U8 hit	Noise8 acc	L5 rob.	L5 hit	L8 hit
Baseline	81.5 ±0.4	60.3 ±0.7	38.6 ±2.8	88.2 ±0.6	8.4 ±0.4	57.1 ±1.8	96.8 ±0.7
FP16	80.9 ±0.8	60.1 ±1.1	39.1 ±3.1	88.2 ±0.6	8.1 ±1.1	57.8 ±2.0	97.1 ±0.5
INT8	83.1 ±1.2	61.8 ±1.4	37.1 ±3.0	88.1 ±0.5	10.4 ±1.4	55.9 ±1.7	96.6 ±0.9
INT4	81.6 ±0.4	58.1 ±1.9	41.8 ±3.4	86.6 ±0.5	8.8 ±0.2	58.8 ±3.0	96.7 ±0.9
Prune 2×	82.4 ±1.8	61.7 ±3.0	36.7 ±2.7	88.2 ±0.5	9.7 ±1.0	55.7 ±1.4	96.1 ±1.4
Prune 4×	82.0 ±1.2	63.5 ±0.3	34.7 ±0.5	88.3 ±0.6	8.5 ±0.2	57.2 ±1.0	97.0 ±0.5
Prune 8×	80.6 ±3.9	60.8 ±2.7	37.8 ±3.6	88.0 ±1.0	8.9 ±1.4	57.5 ±2.1	96.8 ±1.0
Dense-small 2×	77.2 ±1.7	58.4 ±2.8	32.2 ±6.6	87.6 ±0.3	6.9 ±0.7	57.9 ±1.9	96.3 ±0.4
Dense-small 4×	75.4 ±2.6	55.6 ±2.1	38.5 ±0.9	85.3 ±0.3	5.5 ±1.6	63.2 ±0.4	97.9 ±0.4
Dense-small 8×	75.3 ±1.8	53.6 ±2.0	38.0 ±4.9	84.4 ±0.8	4.8 ±0.6	64.7 ±2.5	97.8 ±0.5
KD 2×	80.5 ±1.4	61.6 ±0.3	33.4 ±2.0	87.1 ±0.9	10.5 ±0.1	52.9 ±0.4	95.7 ±0.2
KD 4×	78.2 ±2.9	58.6 ±3.0	38.0 ±2.4	86.7 ±0.5	7.9 ±2.6	56.7 ±2.3	96.5 ±0.6
KD 8×	73.7 ±3.0	55.7 ±1.8	37.0 ±1.5	85.3 ±0.7	5.9 ±1.0	56.2 ±2.8	96.1 ±0.7

U5 / U8 acc = accuracy under the universal 5 / 8 px patch (mean over 10 targets, full test set). U8 hit = target hit rate. Noise8 = random 8 px patch. L5 rob. = LaVAN 5 px untargeted robust accuracy. L5 / L8 hit = LaVAN targeted success. Quantized models: stronger of straight-through and FP32-surrogate attacks.

Table A3. Transfer between full (F) and compressed (C) models (RQ2)

Model	U8 F→C	U8 C→F	L5 F→C	L5 C→F	L5 noise
FP16	38.6 ±2.8	39.1 ±3.1	99.0 ±0.2	99.1 ±0.5	6.2 ±0.9
INT8	35.4 ±3.6	38.1 ±2.7	53.3 ±2.4	89.5 ±1.4	5.9 ±0.7
INT4	34.6 ±3.9	36.1 ±1.7	45.4 ±1.9	32.5 ±1.2	7.1 ±1.1
Prune 2×	29.2 ±2.7	31.4 ±3.8	23.7 ±2.4	29.8 ±1.5	5.5 ±0.9
Prune 4×	29.4 ±2.5	31.0 ±1.8	28.0 ±2.9	28.7 ±2.6	5.6 ±1.1
Prune 8×	30.9 ±2.2	32.6 ±3.6	29.8 ±2.7	26.1 ±4.7	5.8 ±0.9
Dense-small 2×	19.4 ±0.2	15.3 ±3.8	17.9 ±0.8	11.5 ±0.9	8.0 ±0.6
Dense-small 4×	21.3 ±3.0	13.4 ±3.0	21.2 ±2.5	9.4 ±1.2	7.6 ±1.6
Dense-small 8×	17.8 ±1.9	13.2 ±2.2	19.5 ±1.6	8.5 ±1.2	8.1 ±1.1
KD 2×	19.5 ±2.2	18.8 ±1.9	18.3 ±1.0	14.5 ±2.4	6.7 ±1.0
KD 4×	19.8 ±1.6	15.5 ±2.3	20.6 ±0.8	11.1 ±1.6	7.1 ±0.2
KD 8×	16.8 ±1.8	12.7 ±1.1	22.7 ±1.6	9.7 ±1.2	9.1 ±1.2

U8 = universal 8 px target hit rate. L5 = LaVAN 5 px untargeted transfer rate. L5 noise = random-noise patch at the same locations on the compressed model (occlusion control).

Table A4. Physical-EOT ablation (universal 8 px, seed 42 only)

Model	Loc. acc	Loc. hit	Phys. acc	Phys. hit
Baseline	61.2	35.3	71.6	21.2
INT8	63.3	33.7	72.8	21.8
Prune 4×	63.2	34.3	70.8	24.6
Dense-small 4×	56.8	37.5	69.1	21.9
KD 4×	61.5	35.2	70.1	22.1

Location-only EOT vs. physical EOT (rotation ±20°, brightness / contrast 0.8–1.2). Physical EOT gives a weaker digital attack, as expected.

Table A5. PatchCleanser certification (RQ3)

	Vanilla		Defended (mask-augmented)		
Model	Cert-5	Cert-8	Clean (PC)	Cert-5	Cert-8
Baseline	11.0 ±0.3	1.9 ±0.2	95.9 ±0.3	71.6 ±1.5	53.6 ±0.7
FP16	11.0 ±0.3	1.9 ±0.2	95.9 ±0.3	71.6 ±1.5	53.6 ±0.7
INT8	11.0 ±0.3	1.9 ±0.1	96.0 ±0.4	71.8 ±1.6	53.8 ±0.6
INT4	10.9 ±0.4	1.7 ±0.3	95.8 ±0.2	71.1 ±0.4	53.7 ±1.2
Prune 2×	10.9 ±0.7	2.0 ±0.4	95.6 ±0.3	71.4 ±1.5	54.9 ±0.8
Prune 4×	11.3 ±0.8	2.0 ±0.3	95.8 ±0.5	71.2 ±0.7	54.2 ±0.9
Prune 8×	10.9 ±0.8	1.9 ±0.6	95.7 ±0.4	70.1 ±0.7	53.8 ±1.0
Dense-small 2×	10.2 ±0.8	1.7 ±0.2	95.4 ±0.3	67.7 ±0.6	50.5 ±1.0
Dense-small 4×	10.6 ±0.3	1.3 ±0.4	95.1 ±0.6	65.2 ±0.6	47.5 ±0.9
Dense-small 8×	8.9 ±1.0	1.1 ±0.3	94.5 ±0.6	61.1 ±0.9	44.0 ±0.5
KD 2×	11.1 ±1.0	2.0 ±0.2	96.1 ±0.4	70.6 ±1.2	53.7 ±0.4
KD 4×	10.3 ±0.4	1.7 ±0.1	95.1 ±0.2	67.8 ±0.8	50.4 ±0.5
KD 8×	8.1 ±0.7	1.0 ±0.2	94.8 ±0.2	63.9 ±1.3	46.5 ±1.2

Cert-5 / Cert-8 = certified accuracy against a 5 / 8 px patch on the 1,000-image subset. Clean (PC) = clean accuracy of PatchCleanser double-masking inference.

Table A6. Paired differences with 95% t-intervals (RQ1, RQ3, RQ4)

Comparison	Clean	U8 acc	L5 rob.	Cert-5 (def)	Cert-8 (def)
FP16 − Baseline	+0.0 [−0.0, +0.0]	−0.2 [−1.9, +1.5]	−0.3 [−2.2, +1.6]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
INT8 − Baseline	+0.0 [−0.1, +0.2]	+1.5 [−0.5, +3.5]	+1.9 [−0.7, +4.6]	+0.2 [−0.0, +0.4]	+0.2 [−0.2, +0.6]
INT4 − Baseline	−0.2 [−0.5, +0.0]	−2.2 [−5.1, +0.7]	+0.3 [−0.5, +1.2]	−0.5 [−3.3, +2.3]	+0.1 [−1.4, +1.6]
Prune 2× − Baseline	+0.1 [−0.2, +0.4]	+1.4 [−5.2, +7.9]	+1.3 [−0.6, +3.1]	**−0.2 [−0.2, −0.2]**	**+1.4 [+0.6, +2.1]**
Prune 4× − Baseline	+0.2 [−0.3, +0.7]	**+3.2 [+0.8, +5.6]**	+0.1 [−0.6, +0.8]	−0.3 [−4.3, +3.7]	+0.6 [−0.7, +1.9]
Prune 8× − Baseline	+0.2 [−0.2, +0.5]	+0.5 [−4.4, +5.3]	+0.5 [−3.0, +3.9]	−1.5 [−4.6, +1.7]	+0.3 [−2.4, +2.9]
Dense-small 2× − Baseline	−0.0 [−0.3, +0.3]	−1.9 [−8.2, +4.3]	−1.5 [−3.3, +0.2]	−3.8 [−8.1, +0.4]	**−3.0 [−4.0, −2.1]**
Dense-small 4× − Baseline	**−0.6 [−0.9, −0.3]**	**−4.7 [−9.3, −0.2]**	−2.9 [−6.1, +0.2]	**−6.4 [−8.6, −4.2]**	**−6.1 [−7.6, −4.6]**
Dense-small 8× − Baseline	**−1.2 [−1.6, −0.8]**	**−6.7 [−12.3, −1.0]**	**−3.6 [−4.5, −2.7]**	**−10.4 [−14.2, −6.7]**	**−9.5 [−10.2, −8.9]**
KD 2× − Baseline	+0.1 [−0.2, +0.5]	+1.3 [−1.2, +3.8]	**+2.1 [+1.0, +3.2]**	−0.9 [−5.9, +4.0]	+0.1 [−1.0, +1.2]
KD 4× − Baseline	−0.3 [−0.7, +0.1]	−1.7 [−7.8, +4.4]	−0.6 [−6.3, +5.2]	−3.8 [−8.1, +0.5]	**−3.2 [−5.8, −0.6]**
KD 8× − Baseline	**−0.8 [−1.5, −0.2]**	**−4.6 [−8.1, −1.2]**	**−2.5 [−4.0, −1.0]**	**−7.7 [−9.3, −6.1]**	**−7.1 [−8.3, −5.8]**
Prune 2× − Dense-small 2×	+0.1 [−0.2, +0.5]	+3.3 [−9.0, +15.6]	**+2.8 [+1.2, +4.4]**	+3.6 [−0.6, +7.9]	**+4.4 [+3.0, +5.8]**
Prune 4× − Dense-small 4×	**+0.8 [+0.1, +1.5]**	**+7.9 [+2.9, +13.0]**	+3.0 [−0.8, +6.9]	**+6.1 [+3.8, +8.4]**	**+6.7 [+3.9, +9.5]**
Prune 8× − Dense-small 8×	**+1.4 [+0.8, +1.9]**	+7.1 [−2.8, +17.1]	**+4.1 [+1.3, +6.8]**	**+9.0 [+8.2, +9.7]**	**+9.8 [+7.4, +12.2]**
KD 2× − Dense-small 2×	+0.1 [−0.5, +0.8]	+3.2 [−3.6, +10.1]	**+3.6 [+2.1, +5.2]**	**+2.9 [+1.5, +4.3]**	**+3.1 [+1.1, +5.2]**
KD 4× − Dense-small 4×	+0.3 [−0.2, +0.8]	+3.0 [−6.3, +12.4]	+2.4 [−1.5, +6.2]	**+2.6 [+0.1, +5.2]**	+2.9 [−0.4, +6.2]
KD 8× − Dense-small 8×	**+0.4 [+0.1, +0.6]**	+2.0 [−2.3, +6.4]	+1.1 [−0.6, +2.8]	+2.8 [−1.5, +7.0]	**+2.5 [+0.6, +4.4]**

Percentage points, paired per seed, 95% t-interval over 3 seeds in brackets. **Bold** = interval excludes zero. Cert columns use the defended zoo.